

Notes: Hilscher, Raviv, and Reis (RFS, 2026)

How Likely Is an Inflation Disaster?

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1 Big picture

1.1 The limitation of expected inflation (e.g., 5y5y swaps)

Long-dated inflation swaps (such as the 5-year-5-year forward inflation rate, “5y5y”) provide an *expected value* of future average inflation. But policy decisions and risk management often depend on *tail risks*: small changes in the *probability of extreme inflation* can be far more informative than small changes in the mean.

1.2 The key object: disaster probabilities over a forward horizon

Let inflation be defined as log price changes:

$$\pi_{t,t+1} \equiv \log P_{t+1} - \log P_t.$$

Let $\bar{\pi}$ be the inflation target and $d > 0$ be the disaster size. For a forward horizon $(T, T + H)$, define average inflation over that period as

$$\pi_{T,T+H} \equiv \frac{1}{H} (\log P_{T+H} - \log P_T).$$

The paper’s target is the **physical (real-world) probability** that forward inflation is far above or below target:

$$\mathbb{P}(\pi_{T,T+H} > \bar{\pi} + d), \quad \mathbb{P}(\pi_{T,T+H} < \bar{\pi} - d). \quad (1)$$

In the empirical work, the main case is **5y5y** ($T = 5, H = 5$), and disasters are defined as $\bar{\pi} \pm d$ with $d \in \{0.02, 0.03\}$ (“disaster” vs “severe disaster”).

2 Roadmap: why option-implied probabilities need three adjustments

Inflation option prices can be used to infer distributions, but three gaps must be bridged:

1. **Inflation adjustment (real vs nominal payoff):** inflation options pay in *nominal* units, so deep-tail payoffs have different real values across inflation states. Converting to real Arrow–Debreu probabilities requires an inflation adjustment.
2. **Horizon adjustment (forward vs cumulative horizons):** the object of interest is the probability of *average inflation over a forward window*, not cumulative inflation from today to T or to $T + H$. Translating cumulative distributions into forward probabilities requires a model of inflation persistence.
3. **Risk adjustment (risk-neutral vs physical):** option prices identify *risk-neutral* probabilities. If inflation disasters are “bad” states with high marginal utility, then risk-neutral tail probabilities overstate physical tail probabilities. A model of inflation tail risk (not a full general equilibrium model) is sufficient to map Q to P in the tails.

3 Section 2: Theory — from options to disaster probabilities

3.1 Intuition in a stylized multi-state setup

The paper first uses a simple, discrete setup to show the three adjustments transparently.

Nominal option pricing. Consider an option that pays \$1 in a disaster state at the next date. Its price equals the (physical) probability of the disaster times the *real* discount factor times an *inflation conversion* (from \$1 nominal to real consumption units). This already shows why nominal and real objects differ in tail states.

Adjustment 1: from “nominal probabilities” to real risk-neutral probabilities. Let $n_d(1)$ be the conventional nominal-probability object built by scaling the option price by the nominal discount factor. The corresponding real risk-neutral probability $q_d(1)$ satisfies (paper’s Equation (2), restated):

$$q_d(1) = n_d(1) \exp(r(1) + \bar{\pi} + d - i(1)) \approx n_d(1) \exp(d). \quad (2)$$

Intuition: in high inflation states the same nominal payoff is worth less in real terms, so nominal-based probabilities understate real tail probabilities.

Adjustment 2: forward (horizon) probabilities. A forward-period disaster probability depends on inflation persistence inside the window. Using only short-dated options (near T) or only long-dated cumulative options (to $T + H$) misses transitions that occur within the window. Forward-starting option data (annual slices inside the 5y5y window) provide information about persistence.

Adjustment 3: physical probabilities. If inflation disasters coincide more often with “bad times” (high marginal utility), then risk-neutral probabilities exceed physical ones. In a rare-disasters style approximation (paper’s Equation (3), restated):

$$q_d(1) \approx [(\tilde{m} - 1)\tilde{p} + 1] p_d, \quad (3)$$

where $\tilde{p}$ is the probability of a consumption/output disaster conditional on an inflation disaster, and $\tilde{m} > 1$ is the marginal-utility multiplier in those joint-disaster states. **Key point:** the paper only needs to price risk in inflation-disaster states, not in every state of the world.

3.2 General setup and Proposition 1: the complete mapping

This subsection gives the paper’s formal result.

3.2.1 Arrow–Debreu inflation securities

Let $p(\pi)$ be the physical density of inflation π at a horizon date. Define the price (in real consumption units) of an Arrow–Debreu security that pays 1 unit of consumption *only* if inflation equals π :

$$b(\pi) = p(\pi)m(\pi), \quad (4)$$

where $m(\pi)$ is the stochastic discount factor averaged over states sharing the same inflation realization.

3.2.2 Risk-neutral Q probabilities

With real rate r over the horizon, the real bond price is e^{-r} , so Q -probabilities are:

$$q(\pi) = b(\pi)e^r.$$

3.2.3 Nominal N probabilities and the inflation adjustment

Inflation options pay in nominal units, so the corresponding state price is $a(\pi) = b(\pi)e^{-\pi}$. Using the nominal rate i , the paper defines nominal probabilities:

$$n(\pi) = b(\pi)e^{i-\pi}.$$

Let break-even inflation be $\pi^e \equiv i - r$. Then:

$$q(\pi) = n(\pi)e^{\pi-\pi^e}. \quad (5)$$

Interpretation: away from the center of the distribution, Q -probabilities differ systematically from conventional nominal-based objects.

3.2.4 Horizon adjustment: forward vs cumulative distributions

Forward average inflation $\pi_{T,T+H}$ depends on the path of annual inflation inside the forward window. Therefore, converting cumulative distributions into forward probabilities requires a model of inflation persistence that links the joint distribution across years to the distribution of the multi-year average. The paper provides a general expression and then implements it with a parsimonious Markov structure.

3.2.5 Proposition 1: disaster probability decomposition

The paper's key result expresses the physical distribution of forward inflation as:

$$p(\pi_{T,T+H}) = \underbrace{n(\pi_{T,T+H})}_{\text{Options data}} \times \underbrace{\exp[(\pi_{T,T+H} - \pi_{T,T+H}^e)H]}_{\text{Inflation / real-payoff factor}} \times \underbrace{(e^{-r_{T,T+H}} m(\pi_{T,T+H}))}_{\text{Risk factor}} \times \underbrace{\mathcal{H}_{T,H}(\pi_{T,T+H})}_{\text{Horizon factor}}, \quad (6)$$

and then forms disaster probabilities by integrating $p(\cdot)$ above/below the disaster thresholds.

Interpretation of the four factors.

- *Options data:* identifies nominal probabilities tied to observed strike sensitivities.
- *Inflation factor:* corrects nominal payoffs into real Arrow–Debreu probabilities; grows with horizon H .
- *Risk factor:* corrects from risk-neutral to physical probabilities using the pricing of inflation tail risk.

- *Horizon factor*: corrects cumulative distributions into forward-window probabilities using inflation persistence.

4 Section 3: Empirical implementation — all models and how they are estimated

4.1 Inflation options data

Inflation options are caps/floors on *average inflation* over a horizon. A call with strike k pays $\max\{e^\pi - k, 0\}$ scaled by $e^{-\pi}$ (so the payoff depends only on inflation), and its price is denoted $a(k)$. Data:

- United States: monthly series from Oct 2009 to Oct 2024.
- Eurozone: monthly series from Jan 2011 to Oct 2024.
- Horizons: mainly 5y and 10y (liquid maturities), plus forward-starting annual slices in years 5–10.

4.2 Inflation adjustment model: extracting Q from option prices

The paper uses a Breeden–Litzenberger style mapping.

4.2.1 Option pricing identity

For a continuum of inflation states, the call price is:

$$a(k) = \int_{\ln k}^{\infty} \left(\frac{e^\pi - k}{e^\pi} \right) b(\pi) d\pi, \quad (7)$$

where $b(\pi)$ are Arrow–Debreu prices.

4.2.2 Recovering the risk-neutral distribution

Differentiating with respect to strike yields:

$$e^r a'(k) = - \int_{\ln k}^{\infty} e^{-\pi} q(\pi) d\pi, \quad (8)$$

and a second derivative provides the Q distribution in terms of option strike curvature:

$$Q(k) = e^r k a''(k). \quad (9)$$

4.2.3 Recovering the conventional N distribution

Analogously, differentiating the call price also yields a direct formula for the nominal probability distribution:

$$N(k) = 1 + e^i a'(k). \quad (10)$$

Interpretation for practice. Deep out-of-the-money strikes identify tail probabilities via curvature. But one must use the *real* distribution $Q(\cdot)$ (not just $N(\cdot)$) to correctly incorporate the real value of nominal payoffs.

4.3 Horizon adjustment model: inflation persistence via a restricted Markov chain

4.3.1 Inflation process with normal movements and disasters

The paper models annual inflation as:

$$\pi_{t+1} = \bar{\pi} + \varepsilon_{t+1} + d_{t+1}^h - d_{t+1}^\ell, \quad (11)$$

where d^h and d^ℓ are rare positive and negative inflation-disaster jumps.

4.3.2 Discretization into bins

Because strikes come in 0.5% steps, inflation is discretized into 8 bins:

$$\pi(i) \in \{\leq -1, (-1, 0], (0, 1], (1, 2], (2, 3], (3, 4], (4, 5], > 5\}.$$

Inflation dynamics are approximated by an 8×8 transition matrix P with strong restrictions implied by:

- rare-entry Poisson-like disasters,
- fast exit from disasters (especially in the U.S.),
- local volatility in normal times,
- mean reversion toward the 2% bin.

4.3.3 Restricted transition matrix (model object)

The transition matrix has six free parameters:

$(p_{dh}, p_{d\ell})$ (enter high/low disasters), (p_h, p_ℓ) (exit disasters), p_{nn} (normal volatility), p_{mr} (mean reversion).

Other transition probabilities are pinned down by symmetry and row-sum-to-one restrictions.

4.3.4 Estimation via GMM

Each month provides three risk-neutral distributions (discretized into 7 CDF points):

$$q(\pi_{0,5}), \quad q(\pi_{0,10}), \quad q(\pi_{5,6}) \approx \cdots \approx q(\pi_{9,10}),$$

giving 21 moments per month. The paper estimates parameters by GMM, matching these distributions. For parsimony:

- p_{mr} and the disaster-exit probabilities (p_ℓ, p_h) are held constant over the sample,
- the disaster-entry probabilities ($p_{d\ell}, p_{dh}$) and p_{nn} vary over time.

Figure 2 (interpretation). Figure 2 plots the time-varying estimates. For the U.S., perceived inflation volatility (p_{nn}) declines pre-pandemic and rises again afterward. Deflation-disaster entry probability is elevated early in the sample and drops after 2012, with a spike in 2020. High-inflation-disaster entry probability rises strongly in the pandemic/post-pandemic period. For the eurozone, deflation-disaster entry probabilities are generally higher and more variable than in the U.S.

4.4 Risk adjustment model: pricing inflation disasters (tail-only risk model)

4.4.1 Joint model of inflation and consumption/output

To map Q -tail probabilities into physical P -probabilities, the paper specifies:

$$\pi_{t+1} = \bar{\pi} + u_{t+1}^\pi + e_{t+1} + d_{t+1}^h - d_{t+1}^\ell, \quad (12)$$

$$\log c_{t+1} = \log c_t + g + u_{t+1}^c + \beta_0 e_{t+1} - \beta_h d_{t+1}^h - \beta_\ell d_{t+1}^\ell. \quad (13)$$

The common shock e_{t+1} drives normal-times comovement between inflation and consumption (parameter β_0), while β_h and β_ℓ govern how consumption drops in inflation-disaster states.

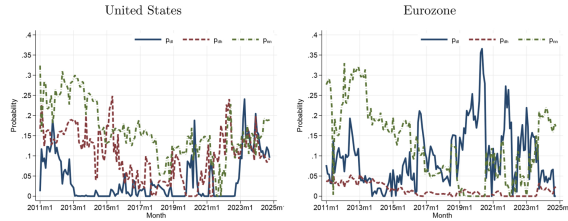


Figure 2
Inflation dynamics: Model parameter estimates
The figure plots time-varying Markov model transition probabilities estimates for the United States and eurozone. We plot the probabilities of moving to a high or low inflation disaster and normal inflation moving.

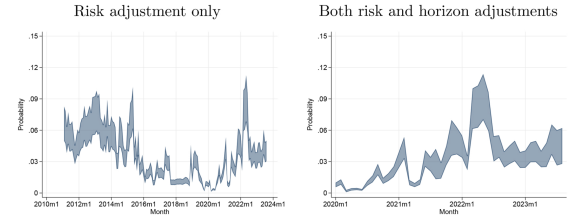


Figure 3
Confidence bands from the adjustments
The left panel shows the 90% confidence band for the US 55y inflation disaster (>4%) probability when standard errors take into account the uncertainty in the risk adjustment estimate; the right panel adds uncertainty in the horizon adjustment.

4.4.2 Pareto tails for disaster sizes

Let the inverse fall in consumption be $z_h \equiv 1/(1 - \beta_h d)$ in a high-inflation disaster. Then z_h follows a Pareto distribution:

$$F(z_h) = 1 - \left(\frac{z_h}{z_{h0}} \right)^{-\alpha_h}, \quad z_h \geq z_{h0} > 1, \quad \alpha_h > 0, \quad (14)$$

and analogously for deflation disasters.

4.4.3 Estimation and calibration

The paper estimates the overlap between inflation disasters and output disasters using historical annual data for advanced economies (1875–2015), and uses standard rare-disasters preference specifications (Epstein–Zin with risk aversion 3) to compute the **risk adjustment factor** (often summarized as a P/Q ratio for tail probabilities).

Table 1**Three inflation disaster probability adjustments***A: High inflation disaster (>4 %) probabilities, 9/21–8/23*

	N_5y	Q_5y	N_10y	Q_10y	Q_5y5y	P_5y5y
U.S., 9/21–8/23, median	20.7%	22.8%	14.0%	17.2%	6.3%	4.2%
U.S., 90th percentile	21.3%	23.8%	23.3%	28.9%	10.7%	7.1%
Eurozone, 90th percentile	14.1%	16.4%	16.2%	20.6%	14.2%	9.4%

B: High inflation disaster probability adjustment factors

	N to Q, 5y	N to Q, 10y	Q, 10y to 5y5y	Q to P, 5y5y
U.S., 9/21–8/23, median	1.09	1.24	0.38	0.66
U.S., median	1.12	1.24	0.40	0.66
Eurozone, median	1.17	1.33	0.92	0.66

C: Deflation (<0%) probabilities, 1/11–12/12

	N_5y	Q_5y	N_10y	Q_10y	Q_5y5y	P_5y5y
U.S., 1/11–12/12, median	6.7%	5.6%	6.9%	4.8%	6.4%	6.2%
U.S., 90th percentile	6.8%	5.9%	6.2%	4.5%	7.5%	7.2%
Eurozone, 90th percentile	11.3%	9.9%	10.5%	7.7%	12.0%	11.6%

D: Deflation probability adjustment factors

	N_5y to Q_5y	N_10y to Q_10y	Q_10y to Q_5y5y	Q_5y5y to P_5y5y
U.S., 1/11–12/12, median	0.84	0.69	1.41	0.96
U.S., median	0.85	0.72	1.33	0.96
Eurozone, median	0.90	0.80	2.29	0.96

Key empirical implication. Risk adjustments differ across tails: high inflation disasters tend to be more strongly associated with deep real downturns than deflation disasters in long-run historical data, so the $Q \rightarrow P$ adjustment is larger for the high-inflation tail.

4.5 Statistical uncertainty (Figure 3)

Two adjustment components are estimated statistically (horizon and risk), so they carry uncertainty. The paper uses:

- bootstrap confidence bands for Pareto tail parameters (risk adjustment),
- asymptotic GMM covariance + delta method for the horizon adjustment,

and Figure 3 shows that estimation uncertainty is modest relative to time variation driven by market prices.

5 Section 4: Table 1 — interpreting the three adjustments quantitatively

5.1 What Table 1 contains

Table 1 summarizes disaster probabilities at different stages:

- N -probabilities (nominal objects) at 5y and 10y horizons,
- Q -probabilities (risk-neutral, real-payoff adjusted) at 5y and 10y,
- Q 5y5y forward probabilities (after the horizon adjustment),

- P 5y5y forward probabilities (after the risk adjustment),
- and corresponding adjustment factors.

5.2 Inflation adjustment (from N to Q)

For **high inflation**, $Q > N$ because nominal payoffs are worth less in real terms in high-inflation states. For **deflation**, the inequality reverses: nominal payoffs are worth more in real terms in deflation states, so $Q < N$.

5.3 Horizon adjustment (from 10y to 5y5y)

This is the largest adjustment in magnitude. For high inflation in the U.S., forward probabilities are much *smaller* than 10y probabilities, consistent with markets expecting mean reversion: inflation spikes are not expected to persist for five more years starting five years ahead. For deflation, especially in the eurozone, forward probabilities can be *higher* than 10y probabilities, consistent with persistent low-inflation concerns.

5.4 Risk adjustment (from Q to P)

Risk-neutral tail probabilities exceed physical tail probabilities when tail states are “bad” for investors. The paper finds a modest adjustment for deflation but a substantially larger one for high inflation, consistent with stronger historical comovement between high inflation episodes and deep real downturns.

5.5 Bottom line in Table 1

After applying all three adjustments, long-horizon disaster probabilities are typically below 5% in normal times, but they move meaningfully around major macro events (deflation fears in the early 2010s, and the 2021–2024 inflation episode).

6 Section 5: interpreting the time series of disaster probabilities (Figures 4–6)

6.1 Deflation fears and unconventional policy (Figure 4)

United States. Figure 4 shows that the probability of a long-run deflation trap is elevated early in the sample but falls below 5% by late 2012 and stays near zero afterward (except for a brief pandemic spike). This implies that long-run deflation-disaster risk in 2011–2014 was likely smaller than suggested by some earlier estimates that focused on near-horizon deflation and/or used different adjustments.

Eurozone. The eurozone exhibits higher and more persistent deflation-disaster probabilities. Unconventional policies (negative rates, QE, forward guidance) are associated with declines in these probabilities, but the declines are not permanent: deflation fears reemerge later in the decade and peak during the pandemic.

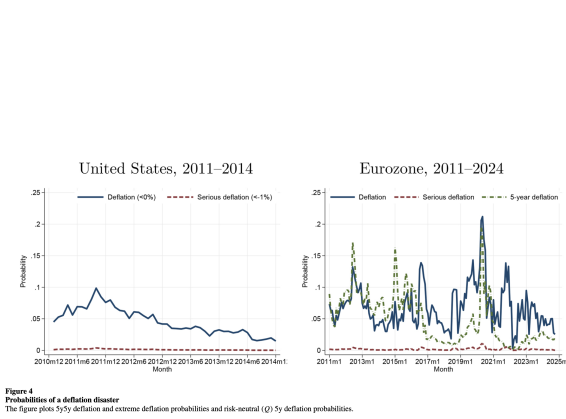


Figure 4
Probabilities of a deflation disaster
The figure plots 5y deflation and extreme deflation probabilities and risk-neutral (Q) 5y deflation probabilities.

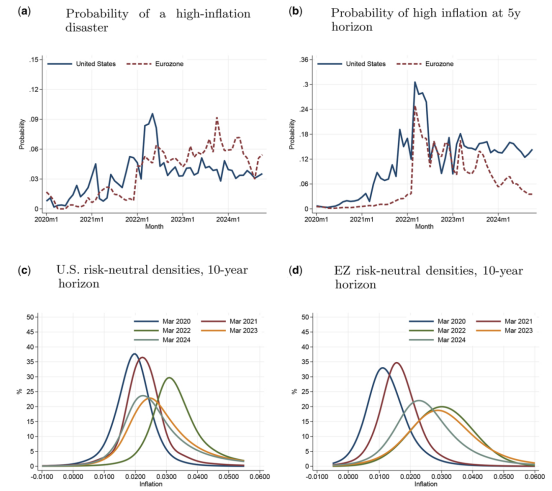


Figure 5
Perceptions of a future inflation disaster during the 2021–2024 inflation disaster
Top panels: 5y (forward) and 5y (near-term) inflation disaster ($> 4\%$) probabilities. Bottom panels: 10y horizon risk-neutral (Q) densities.

6.2 The 2021–2024 inflation episode: deanchoring, reanchoring, scars (Figure 5)

Figure 5 shows three stages:

1. **Pre-2021:** low high-inflation-disaster probabilities in both regions.
2. **2021–mid 2022:** sharp rise in U.S. high-inflation tail probabilities (and later in the eurozone), consistent with deanchoring during the inflation surge.
3. **Policy tightening and after:** probabilities fall as policy tightens, but do not revert fully to pre-2021 levels, indicating “scarring” in tail beliefs.

The figure also highlights that much of the action is in the *tails and skewness* of the risk-neutral distribution, not just in its mean.

6.3 Conditional anchoring: sensitivity to realized inflation (Figure 6)

Figure 6 constructs counterfactual disaster probabilities under alternative current inflation scenarios and alternative near-term inflation paths.

United States. Counterfactual probabilities are close to the baseline: long-horizon disaster risk is relatively insensitive to current inflation realizations, suggesting a stronger perception that policy will prevent persistent future inflation disasters.

Eurozone. Counterfactuals differ more: disaster probabilities would be much lower if current inflation were lower, especially before mid-2023. This indicates that market beliefs about eurozone long-run inflation are more sensitive to current inflation and near-term persistence, consistent with weaker anchoring relative to the U.S.

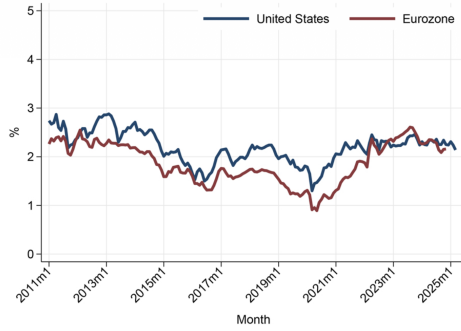


Figure 1
Expected 5y5y inflation

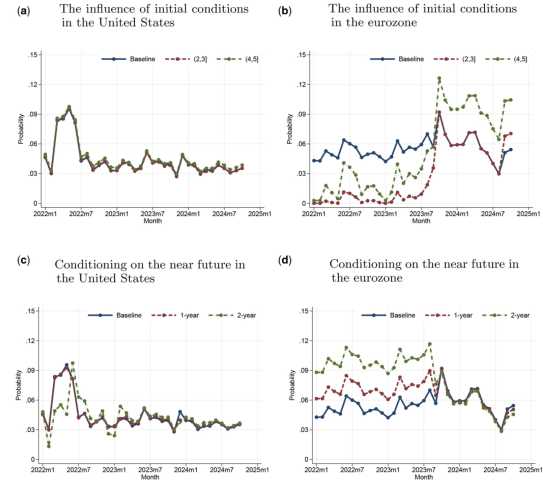


Figure 6
The conditional anchoring of expectations
The figure reports various conditional 5y5y inflation disaster (> 4%) probabilities. Top row: Baseline, based on actual current inflation, and varying current inflation (over the previous year) to being in different ranges, either 2%–3% or 3%–4%; bottom row: Changing, in addition, inflation over the next 2 years.

7 Short summary

1. **Option-implied “probabilities” are not directly physical probabilities.** Always separate N , Q , and P .
2. **The inflation (real payoff) adjustment is mechanically important in the tails and grows with horizon.**
3. **Forward-period disaster probabilities require a persistence model (horizon adjustment).**
4. **Pricing the tail (risk adjustment) is enough.** A full SDF model is unnecessary if the goal is only disaster probabilities.
5. **Empirically:** U.S. long-run deflation-trap risk was smaller and shorter-lived than commonly suggested; eurozone deflation fears were more persistent; and the 2021–2024 inflation episode produced temporary deanchoring and lasting scars, with stronger conditional anchoring in the U.S. than in the eurozone.

8 Conclusion

8.1 Summary

This paper concludes that **long-run inflation tail risks can be measured from inflation options, but only after three conceptually distinct and quantitatively large adjustments.** The object of interest is not just the mean of long-run expected inflation (e.g., the 5y5y swap), but the **probability of persistent inflation disasters** in a distant forward window (e.g., average inflation in years 5–10).

The paper’s central empirical conclusion is that, once the three adjustments are applied to U.S. and eurozone option data (2009–October 2024), the time series of perceived tail risks implies:

- (i) **U.S. deflation-trap fears in 2011–2014 were overstated in conventional readings of market data.** After applying the full correction, deflation-disaster probabilities in the U.S. were *low and short-lived*.
- (ii) **Eurozone deflation-disaster fears were more persistent, and ECB unconventional policies only provided temporary relief.** Deflation tail risk falls around major ECB interventions, but it does not disappear permanently.
- (iii) **Inflation expectations deanchored sharply in 2021–mid 2022.** High-inflation-disaster probabilities rose materially during the inflation surge.
- (iv) **Expectations reanchored as monetary policy tightened.** As policy became restrictive and inflation declined, the high-inflation tail retreated.
- (v) **But the 2021–2024 inflation episode left “scars”.** Even by the end of the sample, the probability of a future high-inflation disaster remains elevated relative to pre-2021 levels.
- (vi) **Long-run inflation beliefs are more tightly anchored in the United States than in the eurozone.** Counterfactual exercises show that temporary inflation shocks (recent or near-term) shift eurozone long-run tail probabilities more than U.S. tail probabilities.

8.2 What is the methodological conclusion? Three adjustments are essential

A second major conclusion is methodological: **conventional option-implied “probabilities” are not the physical tail probabilities policymakers care about.** To get meaningful disaster probabilities, the paper introduces and implements three adjustments:

- (a) **Inflation (real-payoff) adjustment:** inflation options have *nominal* payoffs. Converting them into real Arrow–Debreu probabilities requires correcting for the fact that the same nominal payoff has different real value in different inflation states. This correction is *especially important in the tails*.
- (b) **Horizon (forward-window) adjustment:** the probability of a disaster over a forward window (e.g., 5y5y) is generally not equal to the probability over a shorter horizon (5y) or a longer cumulative horizon (10y). Mapping horizon distributions into forward-window distributions requires a model of inflation persistence.
- (c) **Risk (physical vs risk-neutral) adjustment:** option prices identify *risk-neutral* tail probabilities. If inflation disasters are bad states (high marginal utility), risk-neutral tails will exaggerate physical tails. Importantly, the paper shows that **risk premia differ across the two tails** (high inflation vs deflation) rather than being uniform.

8.3 Quantitative implication: the adjustments are economically large (one clean example)

A key quantitative message (illustrated in the paper’s Table 1) is that the adjustments are not cosmetic. For example, during the U.S. high-inflation period (9/2021–8/2023), the median probability of a *high inflation disaster* (average inflation > 4%) changes dramatically depending on which object one computes:

$$\begin{array}{ccccccc}
 \underbrace{N_{10y}}_{\text{nominal, conventional}} & \approx 14.0\% & \Rightarrow & \underbrace{Q_{10y}}_{\text{inflation-adjusted}} & \approx 17.2\% & \Rightarrow & \underbrace{Q_{5y5y}}_{\text{forward (horizon-adjusted)}} \approx 6.3\%, \\
 & & & & & & \\
 & & & \underbrace{Q_{5y5y}}_{\text{forward (horizon-adjusted)}} \approx 6.3\% & \Rightarrow & \underbrace{P_{5y5y}}_{\text{physical (risk-adjusted)}} \approx 4.2\%. &
 \end{array}$$

This single sequence captures the paper’s main practical warning: **headline “option-implied probabilities” can be very misleading for forward inflation disasters.**

8.4 Why these three adjustments matter

1) Inflation adjustment (nominal payoffs → real probabilities). A deep-out-of-the-money inflation option pays in nominal units. In a high inflation state, a nominal payoff is worth less in real terms, so the option price implies a *higher* real risk-neutral probability than a naive nominal scaling would suggest. Conversely, in deflation states, nominal payoffs are worth more in real terms, so naive nominal scaling *overstates* real risk-neutral deflation probabilities. Hence, the correction changes tail probabilities in opposite directions across tails.

2) Horizon adjustment (cumulative horizon → forward window). The forward-window disaster probability depends on inflation persistence. If inflation is sluggish/mean-reverting, a 10-year average inflation disaster can occur because inflation is high in early years but returns toward target later, so the 5y5y probability can be much lower than the 10y probability. If inflation is persistent (especially for low inflation/deflation concerns), the forward probability can instead be higher than the 10y probability. This is why the paper uses forward-starting distributions inside the window to discipline persistence.

3) Risk adjustment (risk-neutral → physical). Risk-neutral probabilities embed the price of insurance. If high inflation disasters tend to coincide with real downturns (high marginal utility), then tail insurance is expensive: risk-neutral tail probabilities exceed physical probabilities. The paper shows this wedge is larger for high inflation than for deflation, consistent with historical evidence that the high-inflation tail is priced more strongly.

8.5 Conceptual takeaway for reading inflation anchoring

The paper’s final conceptual takeaway is that **“anchoring” should be assessed using tail probabilities, not only the mean.** A stable 5y5y mean can coexist with meaningful time variation in tail risks. Inflation options allow policymakers and researchers to monitor:

- deanchoring (tail probabilities rise sharply),
- reanchoring (tail probabilities decline with policy action),
- scarring (tail probabilities remain elevated after the episode).

8.6 Takeaway

Inflation options can reveal the market's perceived probability of persistent high- or low-inflation disasters at long horizons, but obtaining meaningful forward physical probabilities requires inflation, horizon, and risk adjustments; applied to 2009–2024 data, the method implies U.S. deflation-trap fears in 2011–2014 were overstated, the eurozone's were more persistent, inflation expectations deanchored in 2021–2022 and reanchored with tightening, yet left lasting scars, with the eurozone more sensitive to realized inflation than the U.S.